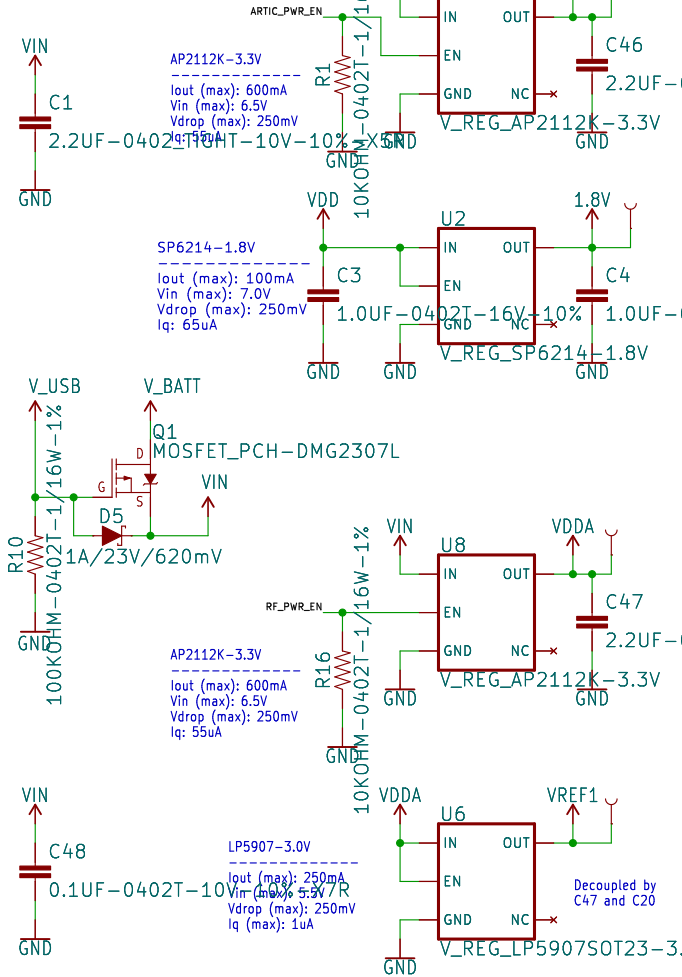
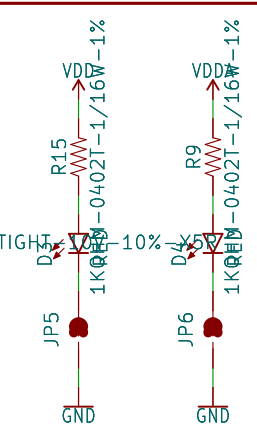


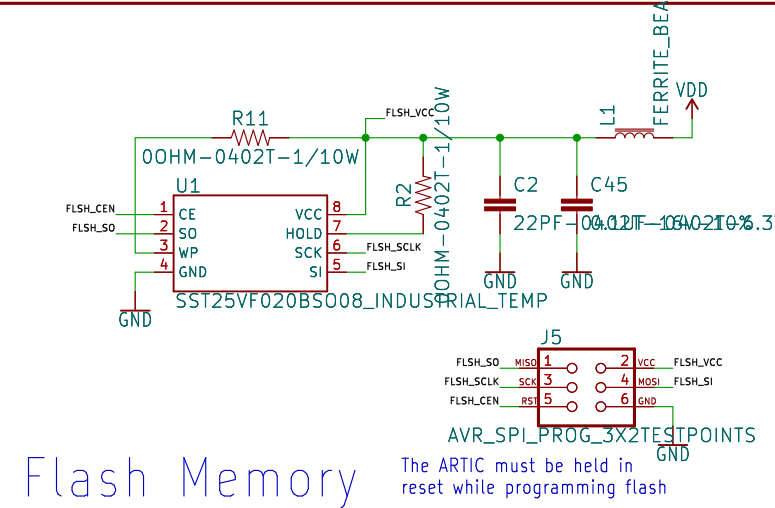
Power Switching & Voltage Regulation



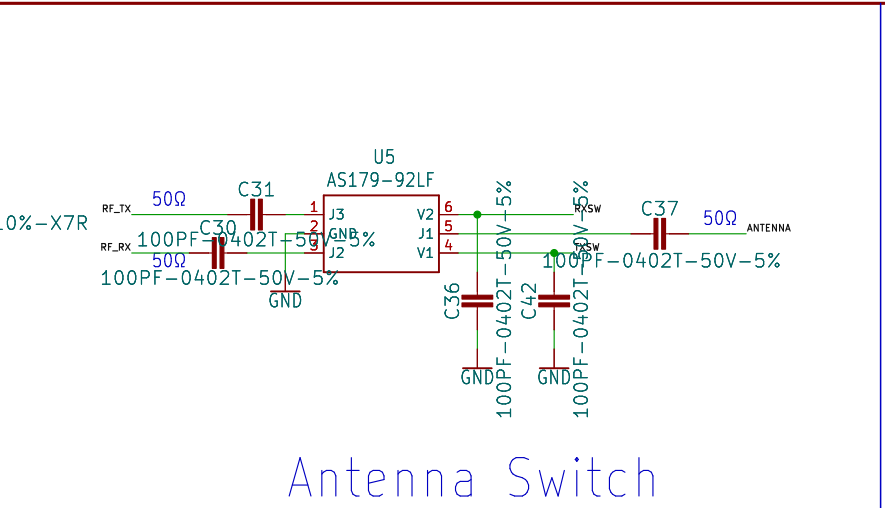
LEDs



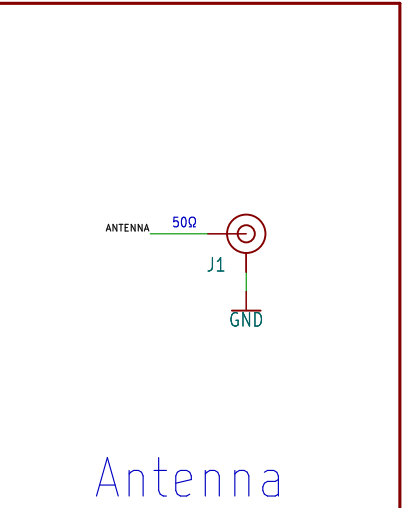
Flash Memory



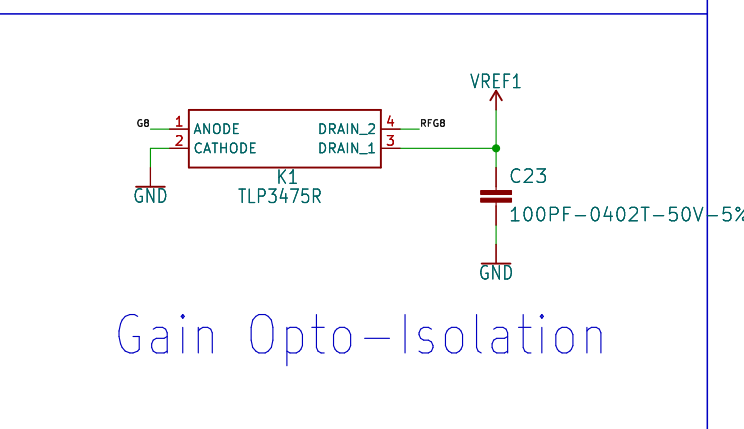
Antenna Switch



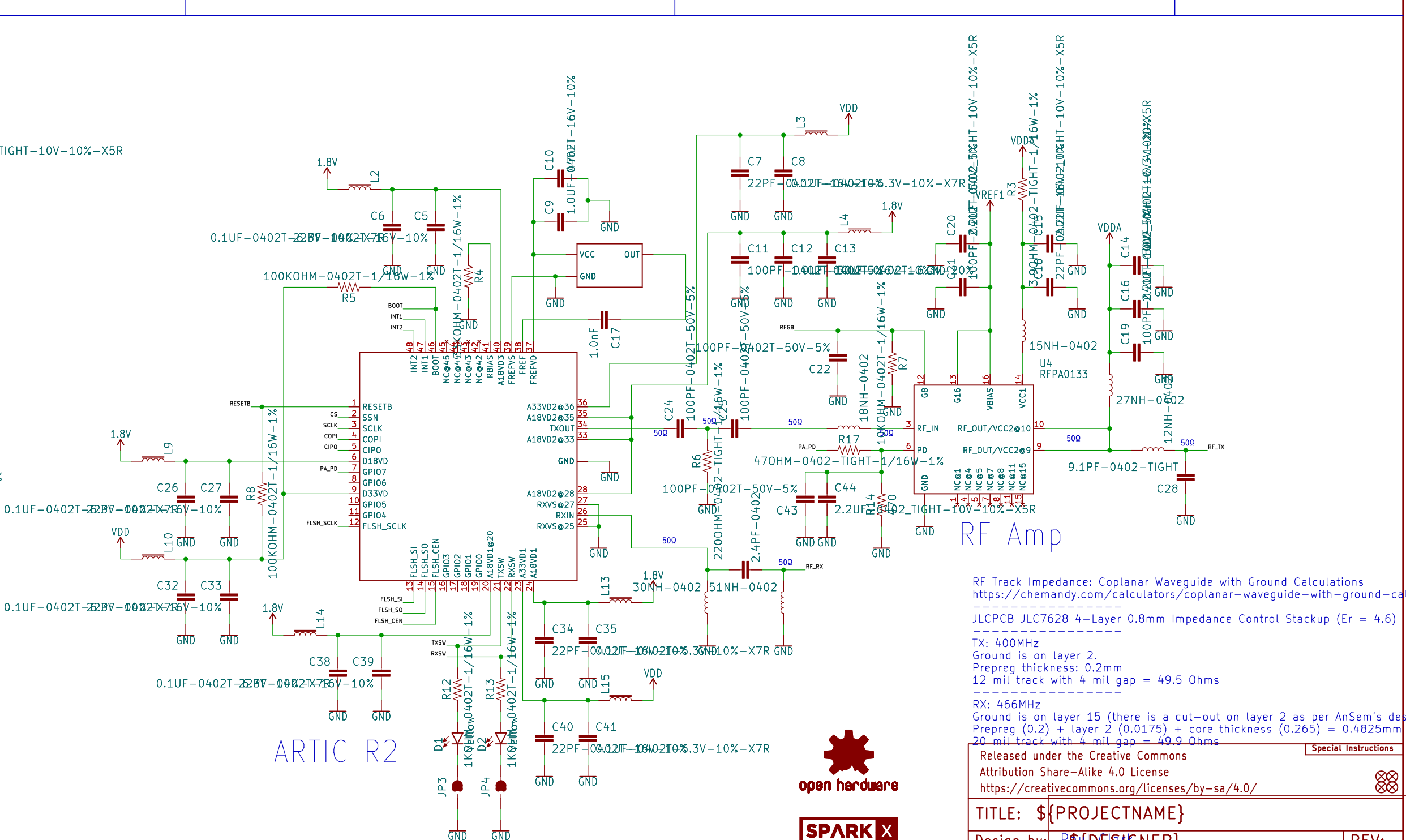
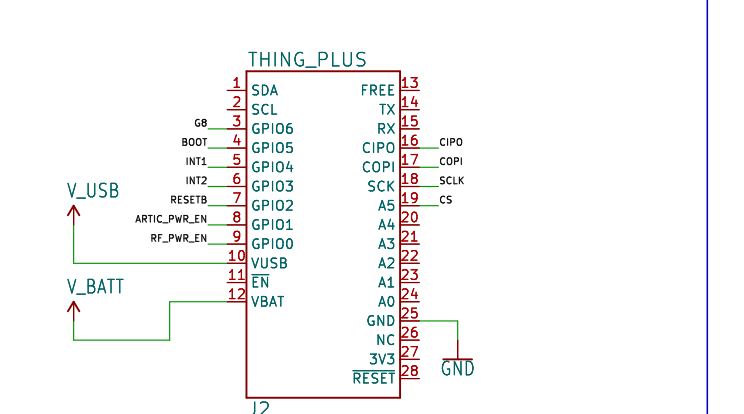
Antenna



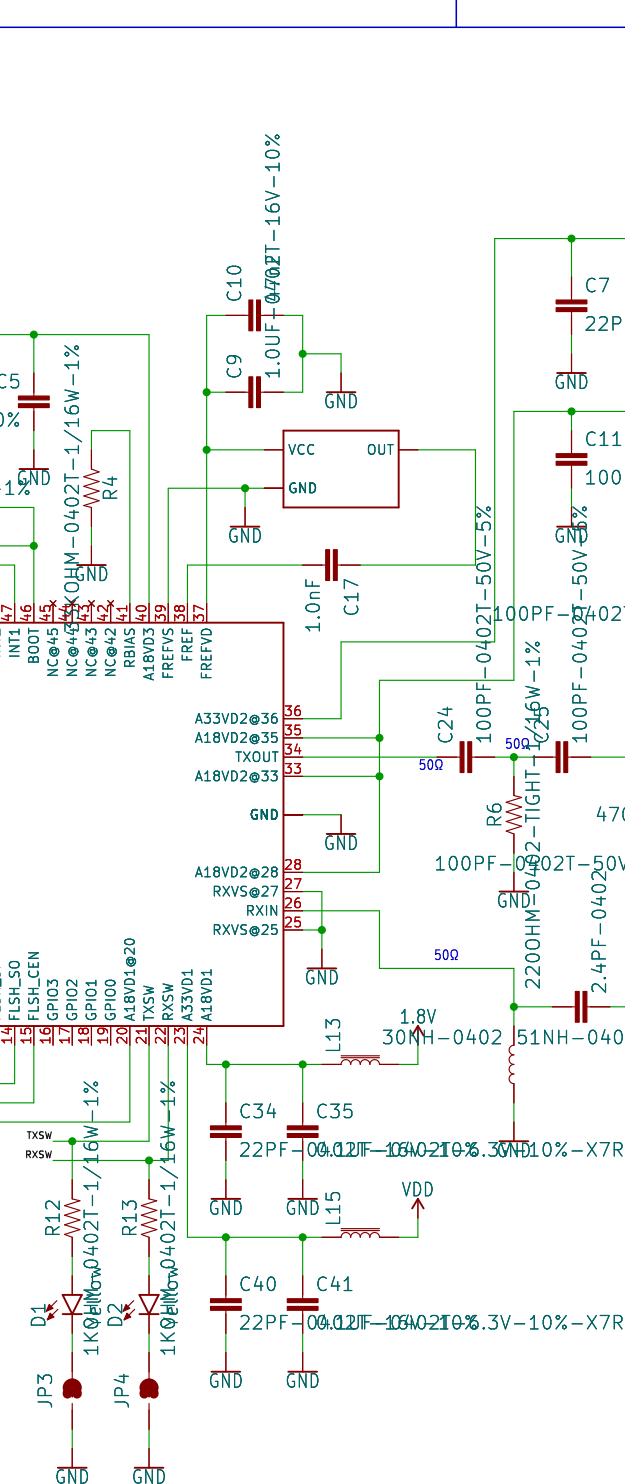
Gain Opto-Isolation



Pins



ARTIC R2



RF Track Impedance: Coplanar Waveguide with Ground Calculations
<https://chemandy.com/calculators/coplanar-waveguide-with-ground-calculator.htm>
JLCPCB JLC7628 4-Layer 0.8mm Impedance Control Stackup (Er = 4.6)
TX: 400MHz
Ground is on layer 2.
Prepreg thickness: 0.2mm
12 mil track with 4 mil gap = 49.5 Ohms
RX: 466MHz
Ground is on layer 15 (there is a cut-out on layer 2 as per AnSem's design guide).
Prepreg (0.2) + layer 2 (0.0175) + core thickness (0.265) = 0.4825mm
20 mil track with 4 mil gap = 49.9 Ohms

Special Instructions	
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TITLE: \${PROJECTNAME}	
Design by: PS\${DESIGNER}	REV: \${REV}
Date: \${CURRENT_DATE}	Sheet: \${#}/\${##}
Size: User	Date:
KiCad E.D.A. kicad-cli 7.0.6-7.0.6-ubuntu22.04.1	Id: 1/1